

LUMBI MUCHUI

Product Leadership, Hardware

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SUMMARY

Senior hardware product manager with **six years** across EV, battery and climate technology, following two years in energy infrastructure. Delivered Sub-Saharan Africa's first locally designed electric motorcycle (**3,500 units and counting**) while cutting unit production cost **53%** to take it from loss-making to break-even; the product earned Roam Electric **2022 Earthshot Prize Finalist** status. Led Africa's first direct-air-capture pilot to first CO₂ capture, and currently own the full hardware portfolio at a battery-swap network across Kenya and Ghana.

EXPERIENCE

Kofa Energy

Senior Product Manager, Hardware

Accra & Nairobi | Sep 2025 – Present

- Own the full hardware portfolio (swap stations, batteries, two-wheel platforms and docking infrastructure) across Kofa's Kenya and Ghana operations, directing engineering, operations, supply chain and external suppliers.
- Architected the next-generation battery platform through parallel dual-sourcing across two qualified assemblers and two cell chemistries (LFP and LMFP), targeting a **20% unit-cost reduction** (–\$150/battery) and **67% cycle-life uplift** (1,500 to 2,500 cycles) while eliminating single-source supply risk.
- Led phased rollout of new powertrain controller firmware across approximately 500 commercial e-motorcycles in Ghana and Kenya, delivering a **25% range increase** and three use-case-tuned operating modes.
- Own New Product Introduction for the next-generation docking hub, taking the programme from initial prototype through customer testing and supplier qualification to a **production-ready state** (EVT through DVT to PVT).
- Lead a continuous-improvement programme across the swap-station network, spanning six workstreams: electrical safety, structural and mechanical integrity, fire and flood safety, connectivity, maintenance operations, and feedback into next-generation hardware design.

Sirona Technologies

Lead Engineer

Brussels, Belgium | Feb 2025 – Aug 2025

- Led installation, commissioning and operations of **Africa's first Direct Air Capture pilot plant** (Lake Elementaita, Kenya; 500 tonnes CO₂/year), achieving first CO₂ capture **two months** from project kickoff.
- Ran capacity and stress-testing campaigns generating the operational data and engineering learnings that fed commercial-scale plant development.
- Took over the commercial-scale DAC plant programme (**30,000 tonnes CO₂/year**, containerised modular architecture) at the DVT gate; drove the plant through PVT to a tested, deployment-ready state at handover.
- Designed the plant-wide instrumentation network (instrumentation cabinetry and harness architecture across the entire DAC stack) including eight automation cabinets built for the commercial plant.

Roam Electric

Product Lead

Nairobi, Kenya | Oct 2023 – Nov 2024

- Took end-to-end ownership of Roam's full product portfolio, four electric-vehicle programmes, with responsibility for product strategy, cost structure and development execution.
- Cut Roam Air motorcycle production cost by **53%** (USD 4,500 to USD 2,100), taking the product **from loss-making to break-even** and widening clean-mobility access across Sub-Saharan Africa.
- Developed technical and product collateral for investor and grant-funding processes, engaging directly with funders.

Technical Product Manager

Nairobi, Kenya | Mar 2022 – Oct 2023

- Owned full New Product Introduction (NPI) cycles (EVT, DVT, PVT, mass production) across four automotive programmes: the Roam Air motorcycle, Roam Move 32-seat bus, Roam Rapid low-floor mass-transit bus, and Griffin electric conversion kits. Directing **30+ engineers** across battery, electrical, mechanical, software and reliability engineering.
- Delivered **Sub-Saharan Africa's first locally designed and developed electric motorcycle**, the Roam Air, now past **3,500 units** and still in production.
- The Roam Air earned Roam Electric **2022 Earthshot Prize Finalist** status (Prince William / The Royal Foundation, top 15 globally) and "World's Most Innovative Companies" recognition.
- Delivered Sub-Saharan Africa's first locally designed and developed electric bus; the Roam Move (32-seater matatu-class), and led product strategy and development for the follow-on Roam Rapid low-floor mass-transit bus.
- Overhauled the product requirements specification process, reducing component manufacturing errors by **75%** over two years (measured via aftersales failure rates and supplier quality improvement).

Mechatronics Engineer, R&D

Nairobi, Kenya | Feb 2020 – Mar 2022

- Designed the core sub-systems of the Roam Air electric motorcycle: battery layout and packaging, high-voltage electrical harness, electronics layout, and mechanical elements.
- Co-developed the Griffin plug-and-play electric conversion kits for Land Cruisers and Land Rover Defenders (30 vehicles installed).

Kenya Pipeline Company

Mechatronics Engineer

Nairobi, Kenya | Sep 2018 – Feb 2019

- Managed pipeline and tank structural-integrity work in the Corrosion Control Department; supported commissioning of the new fuel line to Nairobi's international airport.

Kenya Electricity Transmission Company (KETRACO)

Mechatronics Engineer

Isinya, Kajiado, Kenya | Jun 2017 – Jun 2018

- Installed, tested and commissioned high-voltage substation equipment for the Eastern Electricity Highway, the regional transmission interconnector linking Ethiopia, Kenya and Tanzania.

EDUCATION

BSc, Mechatronics Engineering

Jomo Kenyatta University | 2014 – 2019

- Projects: design and control of a double inverted pendulum; design and fabrication of assistive devices for persons living with disabilities.

Diploma, Programming

Graffins College, Kenya | 2013 – 2014

- Languages: C++, Assembly, Python, JavaScript.

SKILLS

Domain: Hardware product management | Battery and energy-storage systems | BESS | New Product Introduction (NPI) | New product launch | Automotive engineering | Programme management | Agile development

Leadership: Cross-functional team leadership | Supplier coordination | Regulatory compliance (UNECE R136/R156, UN38.3, EU 168/2013) | Full gate ownership (EVT – DVT – PVT – MP) | DFM | DFA | Stakeholder management | Systems thinking

Languages: English (full professional) | Kiswahili (full professional) | French (beginner)